

PHILOSOPHY 130: PHILOSOPHY OF SPACE AND TIME
ASSIGNMENT 3: NEWTONIAN SPACETIME

Due: April 21, in class

Name: _____

Reading: Newton, selections from *De Gravitatione* and the Scholium (C 4); DiSalle, “Newton’s Philosophical Analysis of Space and Time” (C 5); Maudlin, Chs. 1–2.

Please write and draw your answers in the space provided.

1. The tension between relativity and inertia. Consider three bodies—*A*, *B*, and *C*—alone in otherwise empty space. At a given moment, *A* and *B* are 10 m apart and *C* is midway between them. All three are moving so that these relative distances remain constant.

- (a) Draw two diagrams showing the trajectories of *A*, *B*, and *C* over time: one in which all three are moving *inertially* (straight-line, constant speed), and one in which *A*, *B* are moving each orbiting around *C* along a circular path (i.e., accelerating). Label each diagram clearly.
- (b) The relative distances among *A*, *B*, *C* are the same in both diagrams. Briefly explain why a relational definition of motion cannot distinguish between your two cases.
- (c) DiSalle (C 5) argues that Newton gives a “philosophical analysis” of what concepts like inertial motion *require*. Briefly state in your own words what DiSalle means by this, as illustrated in this example.

2. Looking ahead: excess structure. In lecture we noted that we can distinguish several different structural features in Newton's account of space and time, and that each of these supports the introduction of specific quantities (such as absolute velocity and acceleration).

- (a) Maudlin (end of Ch. 2) distinguishes the spatial and temporal structure that Newton's laws *actually use* from additional structure that is not needed. In the table below, place each item in the appropriate column.

Required by Newton's Laws	Excess (not required)

Items: absolute position; absolute velocity; absolute acceleration; the distinction between straight and curved trajectories (in spacetime); absolute time (duration between events); simultaneity.

- (b) Briefly explain why the items you placed in the "excess" column are problematic for Newton, and what kind of argument (from Leibniz or others) targets them specifically.